



1. Project Title & SDG Alignment

Varman: an offline-first response network that lets verified rescuers find each other and match resources when the phone network fails, reaches remote villages through a shared SOS beacon, and turns flood forecasts into early action for the public.

Primary: SDG 11, Sustainable Cities and Communities, Target 11.5 (cut deaths and people affected by disasters, protecting the poor and vulnerable). The first 24 to 72 hours decide who survives, and response fails at coordination, not only at forecasting. **Supporting:** SDG 13 (13.1, resilience to climate hazards), SDG 3 (3.d, early warning and risk management), SDG 9 (9.1, resilient infrastructure).

2. Target Persona & Context

Primary: field rescue coordinators and resource-holding units (SDRF and NDRF teams, fire and emergency services, medical teams, NGO relief camps) working where cellular and VHF links have failed or are overloaded. Each unit is verified by the control room before it can see or accept requests.

Secondary: residents of flood-prone areas, who get alerts, safe places and a one-tap SOS but cannot accept or coordinate requests.

Tertiary: residents of remote, low-connectivity villages with few individual smartphones or devices — reached through one or two shared **Lighthouse SOS beacons** kept at a trusted public location (a school, clinic or panchayat office). Send-only: like the public app, a village beacon can raise an SOS but never accepts or coordinates requests.

Context: India. Built and demonstrated for floods first (urban flooding, river basins). The same ingest layer extends to landslide (GSI), cyclone (IMD) and air-quality (CPCB) alerts.

3. Core Problem Statement

Before the flood: flood, landslide, cyclone and air-quality warnings each come from a different agency (CWC, GSI, IMD, CPCB). None of them is linked to the people who hold the boats, medics and food, so getting help into place is done by hand and usually too late.

During it: the phone network goes down. VHF radios need trained operators, and everything goes through one control room that is already swamped. A team that needs a medical kit, a boat or food has no quick way to find the nearest unit that has one.

Reaching help: even where the mesh and app exist, someone in a remote village with no smartphone and no individual device has no way to raise an SOS at all — the gap a shared village beacon is built to close.

The exact bottleneck: in the first 24 to 72 hours, with no network and a jammed control room, nobody can quickly find the nearest unit that has what they need — and some communities cannot even signal that they need it.

What is built now (prototype)

Varman PWA (React, Firebase, Leaflet): verified rescuer accounts, live request map, Pending / Accepted / Resolved states, safe places, an explainable flood-risk panel (elevation, 30-year rain history, forecast, river discharge), a one-tap **"I'm safe"** button that clears a resolved concern without needing a rescuer to act, per-unit **resource inventories with rough stock counts** (so escalation can say "nearest unit with an actual boat," not just a tag), automatic **duplicate-request flagging** to prevent double-dispatch, a one-click **PDF incident log** for the control room, a no-login public safety map, support for **six languages** (English, Hindi, Tamil, Telugu, Kannada, Bengali), and a virtual field device that simulates LoRa preset broadcasts. The field device, wearable, village beacon, mesh and gateway described overleaf are the proposed design, simulated in this prototype.

Try it out now live: <https://dhriti-e7949.web.app/>

4. Technical Solution Architecture

Orange: proposed hardware, simulated in the prototype. Green / navy: built.

1. Ingest and Risk Engine (cloud)

Rain and river-flow forecasts, elevation and 30 years of heavy-rain history for each monitored place are scored into Safe, Watch, Warning or Danger, with the reasons shown. **Prototype (live)**: plain, explainable rules (IMD rain classes, river flow against the typical yearly peak, low ground, past frequency) over Open-Meteo data (ERA5, GloFAS, Copernicus DEM). **Production (proposed)**: two independent models — an explainable Random Forest trained on CWC / India-WRIS historical flood records, and a separate statistical model built in-house — cross-check each other. Official CWC alerts cover a whole village or district, not a precise point, so for each monitored location the system tracks which of the two models has scored more accurately against that area's own historical flood record and uses it for the local estimate. An official CWC "no flood" is set aside only when both independent models agree a flood is likely — a deliberately strict bar for ever overriding an official all-clear. Live CWC / IMD / INCOIS alerts come via NDMA's Sachet CAP feed; GSI and CPCB feeds extend this to landslide and air quality.



alerts + tips + pre-position

2. Public App (PWA)

Hazard level and reasons, nearest safe places with ground level, safety tips, one-tap preset SOS with GPS, and a one-tap "I'm safe" to clear a resolved concern. Works offline; an SOS made offline queues until signal returns.

3. Rescuer App (verified)

Identity checked by the control room. Live map, accept and resolve requests, capability tags with rough resource counts, mark safe places, duplicate-request warnings.



preset + GPS + vitals

4. Field Device (rescuers)

ESP32, LoRa SX1276, GPS and preset buttons: trapped, medical, food, shelter, missing person. No typing, no spam. **Proposed as a wearable**, adding a heart-rate / stress sensor so a downed or distressed rescuer's own vitals reach the mesh automatically.

5. Lighthouse Device (villages)

One or two fixed, solar-backed units per remote village — same ESP32 + LoRa + GPS core — kept at a school, clinic or panchayat office. Public, send-only: raises a preset SOS into the mesh but, like the app, never accepts or coordinates requests.



stored + relayed

6. Mesh Routing — Four Phases

Phase 1, Flooding: a request from a field device, wearable or village beacon floods out to every nearby node with a message ID (no repeats), a hop limit and an expiry, each hop remembering which neighbour it heard from, until a unit whose tag matches (medical, boat, food) accepts. **Phase 2, Dynamic Source Routing**: on accept, the reply is sent back along that exact recorded chain, hop by hop, instead of flooding again. **Phase 3, Opportunistic Upstream Routing**: in parallel, any node on the path that regains signal syncs the transaction to the control-room database (see Gateway Node, below). **Phase 4, Geographic Routing**: if the rescuer needs more resources, a logistics request is forwarded, hop by hop, toward whichever neighbour is geographically closest to the nearest known gateway, reaching the control room for supply dispatch.



sync

7. Gateway Node

Any node that gets signal, even briefly, syncs its stored requests and their state to the cloud.



sync

8. Cloud & Control-room Dashboard

One live map of every request, from the mesh, the village beacons and public SOS. Resource registry with last-seen and rough stock counts, escalation of requests nobody accepts within N minutes (cross-checked against which verified units report the needed resource in stock), rescuer verification, and flood-status control. Pending, Accepted, Resolved is one state machine for all sources. A one-click PDF incident log exports the full session for accountability.

Why a dedicated LoRa device, not just an app

Phones fail first. In a flood, towers lose power or backhaul or are jammed by traffic, so anything that needs the internet fails exactly when it is needed. **Range.** LoRa relays reach kilometres per hop; phone Bluetooth relays reach about 100 m and only work if many people already run the app. **Presets, not typing.** Buttons replace typing with gloves or shaking hands and stop free-text spam on a slow link. Phones stay the interface for the public wherever signal exists.

BULK PRODUCTION BILL OF MATERIALS (BOM) - 10,000+ UNITS

COMPONENT CATEGORY	SPECIFICATIONS & DETAILS	ESTIMATED UNIT COST (INR ₹)	ESTIMATED UNIT COST (USD \$)
 Microcontroller	ESP32-C3 SoC (System-on-Chip), flash, crystal	₹125	\$1.30
 LoRa Module	Semtech SX1262 Transceiver IC	₹165	\$1.70
 GPS / GNSS Module	Low-power GNSS receiver chip	₹210	\$2.15
 Display & Buttons	0.96" Monochrome OLED panel & Tactile switches	₹95	\$1.00
 Antennas	Internal GPS patch + LoRa whip antenna	₹60	\$0.60
 Power Mgmt & PCB	Custom PCB, PMIC, regulators, load switches	₹215	\$2.20
 Waterproof Enclosure	Custom injection-molded shell with silicone seal	₹95	\$1.00
 Assembly & Testing	PCBA, SMT, Automated RF calibration & QC	₹150	\$1.55
TOTAL ESTIMATED FACTORY COST	Excluding Shipping, Taxes, & Startup NRE	₹1,200	\$14.40

*Note: Prices are estimates for bulk manufacturing (10k+ scale) and subject to market fluctuation.

Data credits: weather and river data from Open-Meteo.com (ERA5, GloFAS), terrain from Copernicus DEM. Alert data from NDMA's Sachet CAP feed (aggregating CWC, IMD, INCOIS, FSI, DGRE). Historical river records for model training from data.gov.in / India-WRIS. Cost figures are component estimates. Rain history and both the Random Forest and statistical models are modelled and advisory; official CWC, IMD and NDMA/Sachet alerts always take precedence.